$dN_p: T_pS \to T_pS$

自待 $dN_p(w_1), w_2$

$M \to S^2$

$T_pM \to T_{N(p)}S^2$ 线性映射

$\to dN_p: T_pS \to T_pS$

(w_1, w_2)

$\langle dN_p(w_1), w_2 \rangle = \langle w_1, dN_p(w_2) \rangle$

$X(u,v)$ $w_1 = X_u$ $w_2 = X_v$ $\alpha(t)$ $\alpha(0) = p$

$dN_p(\alpha'(0)) = \frac{d}{dt}(N(u(t), v(t))) = N_u \cdot u'(0) + N_v \cdot v'(0)$

$dN_p(X_u) = N_u$

$dN_p(X_v) = N_v$

$\langle N_u, X_v \rangle = \langle X_u, N_v \rangle$

$\langle N, X_u \rangle = 0 \Rightarrow \langle N, X_{uv} \rangle + \langle N_v, X_u \rangle = 0$

$\langle N, X_v \rangle = 0$

$Q(v) = \langle dN_p(v), v \rangle$

$\Rightarrow \mathrm{II}_p(v) = -Q(v)$

$\alpha(s)$ $\alpha(0) = p$. $N(s)$ $\langle N(s), \alpha'(s) \rangle = 0 \Rightarrow \langle N(s), \alpha'' \rangle = -\langle N', \alpha' \rangle$

$\alpha'(0) \to dN_p \to N'(0)$ 在0处 $\langle N(0), \alpha''(0) \rangle = -\langle dN_p(\alpha'), \alpha' \rangle$

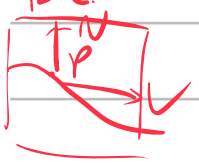
$|\alpha'| = k$.

(4) $\cos\theta = \langle n, N \rangle$

$k_n = k\cos\theta$

$\langle N(0), \alpha''(0) \rangle = \langle N, k_n \rangle = k\langle N, n \rangle = k\cos\theta = k_n$ → 法曲率

推论.



$z = y\varphi$ $(0,0,0)$ $dN_p = 0$ ← $k_n = 0$ ← $y(x) = x^4$ 在 $x=0$ $k=0$ $(x, x^4) \to (0, \varphi x^3)$

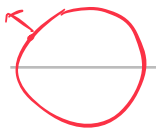
$\to (0, 12x^4)$

$= 0$



S^2 $I_p(v)$

$k = 1$.



↙ 法曲率 $|\kappa(-1)| = 1$ $I_p(v) = -1$

$dN_p \to T_pM$ e_1, e_2. $dN_p(e_1) = -k_1 e_1$

$dN_p(e_2) = -k_2 e_2$

$v = e_1\cos\theta + e_2\sin\theta$ $|v| = 1$

$k_n = -\langle dN_p(e_1\cos\theta + \dots), \dots \rangle$

$= k_1\cos^2\theta + k_2\sin^2\theta$

dN_p 行列式 $\det dN_p = K$

$\operatorname{trace} dN_p = -H$.

① 曲率线

脐点 $k_1 = k_2$

$X(u,v)$ $N=\frac{X_u \wedge X_v}{|X_u \wedge X_v|}$ $\alpha(t)$

$\alpha(t)$ 在 p 处 $\alpha' = X_u u' + X_v v'$

$dN_p(\alpha') = N_u u' + N_v v'$

$N_u = a_{11} X_u + a_{21} X_v$

$N_v = a_{12} X_u + a_{22} X_v$

$dN_p(\alpha') = (a_{11}u' + a_{12}v')X_u + (a_{21}u' + a_{22}v')X$

$dN_p\begin{pmatrix} u' \\ v' \end{pmatrix} = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}\begin{pmatrix} u' \\ v' \end{pmatrix}$

$e = \langle N, X_{uu} \rangle = -\langle N_u, X_u \rangle$

$f = \langle N, X_{uv} \rangle = _ _ .$

$g = \langle N, X_{vv} \rangle = _ _ _$

$-\begin{pmatrix} e & f \\ f & g \end{pmatrix} = A\begin{pmatrix} E & F \\ F & G \end{pmatrix}$. $A = \begin{pmatrix} a_{11} & a_{21} \\ a_{12} & a_{22} \end{pmatrix}$

$$A = -\begin{pmatrix} e & f \\ f & g \end{pmatrix}\begin{pmatrix} E & F \\ F & G \end{pmatrix}$$

$a_{11} = \frac{fF - eG}{EG - F^2}$. $_ _$. $_ _ _$

$K = \det(a_{ij}) = \frac{eg - f^2}{EG - F^2}$

$-k_1, -k_2$ dN_p.

$dN + kI$ 不可逆. $k = k_1, k_2$

$\det\begin{pmatrix} a_{11}+k & a_{12} \\ a_{21} & a_{22}+k \end{pmatrix} = 0$ $k = k_1, k_2$.

$k^2 + k(a_{11} + a_{22}) + \det(a_{ij}) = 0$

$a_{11} + a_{22} = -(k_1 + k_2)$ $H = \frac{1}{2}(k_1 + k_2)$

$a_{11} = \frac{fF - eG}{EG - F^2}$ $a_{22} = _ _ .$ $H = \frac{1}{2}\frac{eG - 2fF + gE}{EG - F^2}$ ✓ $\Rightarrow k^2 - 2Hk + K^2 = 0$

· 全脐

S. 全脐 → 平面/Ball.

$dN(w) = \lambda(q) w$.

$\lambda(q)$ local. $X(u,v)$ $N_u a_1 + N_v a_2 = \lambda(q)(X_u a_1 + X_v a_2) \in \omega$

由 a_i 任意 $\Rightarrow N_u = \lambda X_u$ $N_{uv} = \lambda_v X_u + \lambda X_{uv}$ 相减 $\lambda_v X_u = \lambda_u X_v$ 得 $\lambda_v = \lambda_u = 0 \Rightarrow$ local const

$N_v = \lambda X_v$ $N_{uv} = \lambda_u X_v + \lambda X_{uv}$

$\lambda \equiv 0 \Rightarrow N \equiv$ Const ✓

$\lambda \equiv A \neq 0$ $p \in M$ $Y(u,v) = X(u,v) - \frac{1}{\lambda} N(u,v)$

$\left| X - \frac{1}{\lambda} N \right|$

$\left[X(u,v) - \frac{1}{\lambda} N(u,v)\right]_u = X_u - \frac{1}{\lambda} N_u = 0$

$\Rightarrow$ Const

$\|X - Y\| = \frac{1}{\lambda}$ (球)

$p \in M$. 都有 Local ✓ T_pM 一例 /22 两例

Pf. $X(u,v)$ $p = X(0,0)$ $N(p)$

$d = \langle X(u,v) - X(0,0), N(p) \rangle$

$X(u,v) = X(0,0) + X_u u + X_v v + \frac{1}{2}(X_{uu} u^2 + 2X_{uv} uv + _) + R$ $R = o(u^2 + v^2)$

$d = (X_u u + X_v v + \frac{1}{2}(__), N(p))$

$= \frac{1}{2}(eu^2 + 2fuv + gv^2) + R_1$.

$= \frac{1}{2} \mathrm{II}_p(w) + R_1$; $w = X_u u + X_v v$

$R_1 = o(|w|^2)$ $d|w|^2 = \frac{1}{2}\left|w|^2 \mathrm{II}_p(w)\right| + o(1) > 0$

$|w| = 1$

Dupin 标线

$p \in M$. T_pM 子集 $\{w \mid \mathrm{II}_p(w) = \pm 1\}$

(e_1, e_2) $w = \rho v$. $|v| = 1$. $\rho =$ radius

$v = e_1 \cos\theta + e_2 \sin\theta$. ($k_1, k_2$)

$\pm 1 = \rho^2 \mathrm{II}_p(v) = k_1 \rho^2 \cos^2\theta + k_2 \rho^2 \sin^2\theta$

$= k_1 x^2 + k_2 y^2$. (x, y) ω e_1, e_2



M. $h(x,y)$ $z=h(x,y)$

$h(0,0)=p$. $h_x=h_y=0$.

$X_u=(1,0,h_x)$ $X_v=$ —— . $N(x,y)=\frac{(-h_x,-h_y,1)}{(1+h_x^2+h_y^2)^{\frac{1}{2}}}$

$X_{uu}=(0,0,h_{xx})$ $e=\frac{h_{xx}}{\sqrt{1+h_x^2+h_y^2}}$.

f,g

Let $v=xX_u+gX_v$.

$\mathrm{II}_p(v)=ex^2+2fxy+gy^2$

$=h_{xx}x^2+2h_{xy}xy+h_{yy}y^2$.

h $(0,0)$ Hesse.

$z=\varepsilon$ $\{(x,y)\mid h(x,y)=\varepsilon>0\}$. 闭线 → C

$f=h_{xy}(0,0)$ $k(p)=-\frac{\langle N_u,X_u\rangle}{\langle X_u,X_u\rangle}=h_{xx}$

$k_2(p)=h_{yy}$

$h(x,y)$ $h(x)=\frac{1}{2}\text{Hesse}+R$

$=\frac{1}{2}(k_1x^2+k_2y^2)+R$

闭线 C $\boxed{k_1x^2+k_2y^2=2\varepsilon.}$

$M\xrightarrow{\omega}\mathbb{R}^2$

向量场 on M

$p\in M$. $\alpha: I\to U$

$\alpha(t)$ $\boxed{\alpha'(t)=\omega(\alpha(t))}$

· lemma

ω 向量场

p $\omega(p)\neq 0$. W. 对于 $q\in W$. $df_q\neq 0$

$\mathbb{R}^2$ $(0,0)$

$w(p) = \overrightarrow{(1,0)}$

$\alpha: U \times I \to W$

$\alpha(q,0) = q$

$\alpha(q,t) =$ 流 t 时间

$\mathcal{Z}: (U \times I) \cap \{(x,y,z) \mid x = 0\}$